Syn2Real: Forgery Classification via Unsupervised Domain Adaptation

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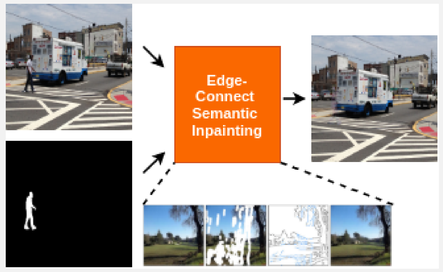
Motivation

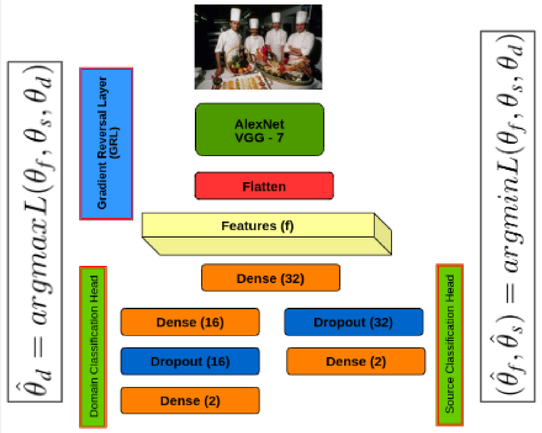
- Copy-Move Forgery (CMF) have significant effects but it's difficult to detect due to its complexity.
- Previous works employed supervised learning algorithms and requires a huge chunks of labeled images. Forged dataset is seldom available for various fields.
- Trained on synthetic images doesn't generalize well on realistic dataset.
- **Domain Adaptation** learns features from source data distribution and use that knowledge to classify images in target data distribution.

Proposed Methodology

Generated a synthetic dataset of 80,000 images using semantic inpainting and CMF techniques on COCO dataset. Adapted from source data(D_s) to target dataset(D_t) using DANN and DDC.

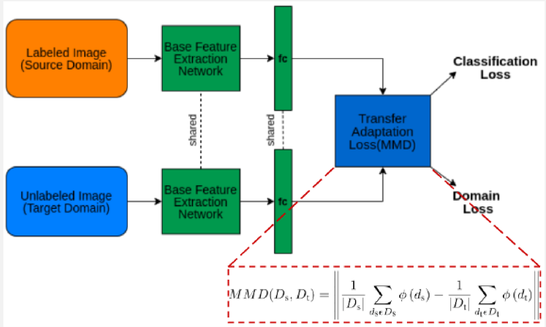


Domain Adversarial Neural Network (DANN)



$$L(\theta_f, \theta_s, \theta_d) = \sum_{x=1\ldots N} L_s(D_s(D_f), y_i) + (\Delta) \sum_{x=1\ldots N} L_d(D_t(D_f), y_i) \quad (1)$$

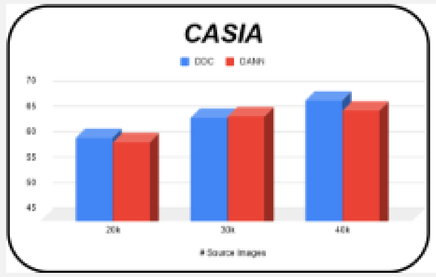
Deep Domain Confusion (DDC)



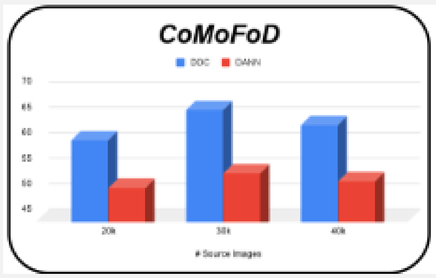
$$L_{\text{total}} = L_{classify}(x, y) + \alpha MMD^2(D_s, D_t) \quad (2)$$

Results & Discussion

- CMF images helped the network to recognize same feature characteristics, whereas, inpainted images make the network analyze edge discrepancies around forged region.



- ***COCO → CASIA:*** With 3,979 images for training, the network is able to generalize for large number of parameters. Evaluating on 996 images, DANN+VGG outperforms other models.



- ***COCO → CoMoFoD:*** Due to presence of only 200 images in total, high number of parameters can't be optimized. Thus, DDC+AlexNet performs best amongst all.

Architecture	P	R	F-1	P	R	F-1
BusterNet[1]	**78.22**	73.89	75.98	57.34	49.39	49.26
DDC+Alex	68.02	97.73	80.18	**65.62**	**97.86**	**78.99**
DDC+VGG	57.27	95.02	70.13	65.39	91.24	76.18
DANN+Alex	66.19	98.47	79.16	51.05	83.5	63.34
DANN+VGG	68.26	**99.84**	**80.34**	49.65	72.04	58.75

Table: P=Precision, R=Recall and F-1 scores on CASIA and CoMoFoD dataset respectively.

Conclusion

- Outlined two unsupervised approaches to detect CMF in new domains.
- Suggested two new methods for synthetic dataset generation.
- Improved the F-1 scores on CASIA and ComoFoD datasets by 4.36% and 29.73%, respectively.
- Outperforms previous approach that incorporates supervised pixel-based learning.

Future Work

1. Explore GANs to generate more realistic images.
2. Incorporate Splicing and inpainting forgery to make our model robust.
3. Improve the precision score.

References

[1] Y. Wu, W. Abd-Almageed, and P. Natarajan. Busternet: Detecting copy-move image forgery with source/target localization. In *ECCV*, Sept. 2018.

[2] Y. Ganin and V. Lempitsky. Unsupervised domain adaptation by backpropagation. In *Volume 37*, ICML'15, pages 1180–1189. JMLR.org, 2015.

[3] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell. Deep domain confusion: Maximizing for domain invariance. *ArXiv*, abs/1412.3474, 2014.

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